



Understanding Joshimath landslide using PS interferometry and PSDS InSAR

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The recent subsidence at Joshimath in the Indian State of Uttarakhand led to the displacement of thousands of residents. Large cracks developed in the buildings and on the ground. No extensive and comprehensive deformation measurement of this event has been made. In this study, we use both PS and PSDS InSAR time series to investigate the magnitude, spatiotemporal as well as kinematic evolution of this slow-moving landslide. Eighty-seven ascending Sentinel-1 scenes with a temporal baseline of 1056 days from 2020 to 2023 were stacked for interferometric analysis. StaMPS is employed to identify PS points by their amplitude and phase information. TomoSAR is utilized to stipulate a coherence matrix to form a dense PSDS network of interferograms to surge point density for suitable phase unwrapping. PS and DS points are coupled to develop slope velocity maps revealing mean displacement rates of -84 mm for PS and -107 mm for PSDS, respectively. Cross-section profiles drawn on the slopes of subsidence show target scatterers on CS1, CS2 and CS4, yield a cumulative displacement of 400 mm in the last 3 years. CS3 and CS5 show a total displacement of about 350 mm. This study applies PSDS time-series InSAR to decipher ground movement in traditionally decohered environments. It also seeks to establish the boundaries and intensity of subsidence to aid in the mitigation of failure progression.

Keywords. InSAR; TomoSAR; landslide; Joshimath; PSI; distributed scatterer.

1. Introduction

Landslide is the movement of mass materials down the slope. This hazard is most extensive and intensive in the mountainous regions due to their high-energy environments. The presence of steep slopes, infirm lithology and structural variability compound the effects of such a hazard. As such, landslides are most prominently observed in the rugged topography of the mountainous regions, especially the Himalayas. Though most landslides are natural hazards, the increased frequency and magnitude of such hazards have been ascribed to

human activity (Gerrard 1994). The progression and severity of human-induced landslide hazards in the Himalayan region have been successfully linked by Ives Jack and Bruno (2001) and famously christened as the Himalayan Environmental Degradation Theory. Barnard *et al.* (2001) accredited about two-thirds of the active landslides in the Garhwal region to human activity. Though later research showed the topographic elevation, slope inclination, aspect, vegetation, lithology, structural aspects, geology as well as climatic controls as the main drivers of mass wasting in the region (Gerrard 1994; Awasthi *et al.* 1995;

Panikkar and Subramanyan 1996; Chauhan *et al.* 2010). Owing to complex topography, tectonic evolution, intense river erosion, rapid uplift and a myriad of other geological and climatic factors, Garhwal Himalayas have seen several disasters of varying magnitudes like extensive landslides, debris slides, rotational slides, slumps, rockfalls, avalanches, and flash floods destroying everything in their path. The area of most interest in the Garhwal Himalayas is the Chamoli district in the Indian state of Uttarakhand, where the prevalence of some natural hazards like debris slides, landslides and cloudbursts have severely affected both life and property in recent memory. In mid-June of 2013, Kedarnath flash floods caused the death of more than 6000 people and unprecedented damage to the property just 50 km west of Joshimath town (Sati and Gahalaut 2013). On February 7, 2021, a huge ice cum rock slurry avalanche was triggered by the detachment of an ice-rock slab from the north-face of Raunthi peak of Trishul mountain range into the Raunti Gad valley just 25 km southwest of the town of Joshimath killing hundreds, destroying everything downstream including Tapovan Vishnugad hydropower plant and completely washing away Dhauliganga Dam (Siddique *et al.* 2022).

In this study, we will examine and quantify the extent and magnitude of concurrent landslides in the town of Joshimath, which has created widespread panic and real-time danger to the residents of the affected town. Initial reports in October 2022 began to pour in, and ultimately, buildings in the township started tilting with a concomitant water efflux from the local gullies and rills (Chadha 2023). Subsequently, in the following weeks, the town saw further deformation and an extensive subsiding in large swathes. In the first week of January 2023, large cracks developed in over 560 houses from all the nine wards of the town (The Indian Express| 09 Jan 2023), which thereafter increased to 868 dwellings (The Indian Express| 15 Feb 2023). A total of 900 members of 269 families were displaced in this event (The Hindul 20 Jan 2023). On 19 February, new fissures began to develop in the juxtaposed highway (The Hindul 20 Feb 2023) ahead of the commencement of Char Dham Yatra (pilgrimage to four ancient Hindu sites in Garhwal Himalaya). Although the town sits on a tectonic boundary, this recent event is deemed aseismic in nature (Chadha 2023).

Kanungo and Sharma (2014) attributed short and intense antecedent rainfall (0.87 m/h) to the increased risk of landslides in the Chamoli region.

Other researchers deduced the slope of hills, lithology as well as copious rainfall as the main environmental controls on landslide occurrence (Mey *et al.* 2023) in the locale. Agarwal *et al.* (2023) identified Joshimath–Kedarnath region as a highly vulnerable hotspot prone to natural hazards owing to high surface radiative temperature by dint of population growth, soaring infrastructure, increased anthropogenic land use and diminishing vegetative cover (Verma *et al.* 2022) as ameliorating factors of the influence of climate change in the region. Certain other events are supposed to have aggravated the already fragile ecosystem, like the construction of Vishnu-Gad Hydro-Electric project, in pursuit of which the Head Race Tunnel (HRT) punctured through a Quartzite water-bearing strata gushing enormous amount of water at the rate of 600–700 L/s for more than a month, some 3 km inward the left bank of Alaknanda near Shelong village (Bisht and Rautela 2010; Nawani 2015). Road construction and subsequent widening were considered as *prima facie* in landslide causation along the National Highway (Saha *et al.* 2002). Barnard *et al.* (2001) performed cosmogenic radionuclide dating of strath terraces along the Alaknanda River and established that human activity has influenced about two-thirds of the active landslides in the region, thereby exacerbating the human-induced denudation.

Since the launch of several SAR satellites three decades ago, radar interferometry has been booming, especially when applied to deformation scenarios, where the need for millimetre-level precision arose. Since quantitative analysis is important for geological interpretation, prognosis, or resolving mechanical and geometrical properties of an active landslide, the application of Interferometric Synthetic Aperture Radar becomes necessary to monitor the surface displacement field. While multiple time-series InSAR methods have been developed since its inception for measuring such deformations, most of them can be categorized into two main approaches. One is known as PS (Persistent Scatterer) technique, where the coherence characteristics of some permanent scatterers dominate and remain constant throughout time in a resolution cell (Ferretti *et al.* 2001; Hooper *et al.* 2004, Hooper 2006) and the other method, SBAS (Small Baseline Subset) involves inverting conventionally derived interferograms and extracting deformation on the basis of grouped scatterer coherence in smaller temporal subsets (Lundgren *et al.* 2001; Berardino *et al.* 2002;

Hooper *et al.* 2012). Persistent Scatterer Interferometry technique provides high-quality information about point targets in an urban setting, but the density of such PS points in a low coherence zone is often sparse. Targets like forests, agricultural fields, bare soil or rock surfaces form such low coherence zones and are termed as distributed scatterers (Even and Schulz 2018). To overcome this issue of point scantity, PSDS technique is wielded, where both PS and DS (figure 1) are exploited in tandem to measure range change while enriching low coherence target with PS points (Ho Tong Minh and Ngo 2022).

Interferometry has been used by several researchers to monitor and quantify landslides in the Himalayan–Alpine belt (Fiorucci *et al.* 2011; Vöge *et al.* 2015; Dong *et al.* 2018; Aslan *et al.* 2020; Bekaert *et al.* 2020; Chen *et al.* 2020; Hussain *et al.* 2021; Zhang *et al.* 2021; Dai *et al.* 2022; Jennifer *et al.* 2022; Lacroix *et al.* 2022; Roy *et al.* 2022; Yao *et al.* 2022). From previous studies, it can be summarized that InSAR is one of the most effective methods for slow-moving landslide observation, interpretation, monitoring and assessment due to its high accuracy, extensive coverage, and long temporal acquisition under all weather conditions (Burgmann *et al.* 2000; Hanssen 2001). In view of the recent Joshimath tragedy and consequent government restriction to site visitation, we apply

PSDS Interferometry to both evaluate the areal extent of landslide-induced subsidence in Joshimath as well as gauge the amount of deformation incumbent on the township.

2. Study area

Joshimath is a small hilly town located in the upper reaches of Alaknanda valley of Garhwal Himalayas, in the disaster-prone Chamoli district of Uttarakhand in India (figure 2a). Settled between the meridians $79^{\circ}32'08''$ – $79^{\circ}35'28''$ E and latitudes of $30^{\circ}31'30''$ – $30^{\circ}34'09''$ N and an approximate area of 20 km^2 , the town shows a steep elevation gain from 1950 m (amsl) at the foot of the slope to 2950 m (amsl) at the top within a short run of 4 km. Alaknanda River (glacier-fed, tributary of river Ganga) flowing in the north–south direction (following the axial trend of Pipalkoti Anticline) confluences with the east–west flowing Dhauliganga at the basal periphery of the town. Annually, the area receives around 1500 mm of precipitation, most of which occurs during the monsoon seas (Devrani and Singh 2014). As per the 2011 Census, a total of 3,898 families were residing in Joshimath town with a total population of 16,709 (Home | Government of India (censusindia.gov.in)). Current population of the town is estimated at 25,000

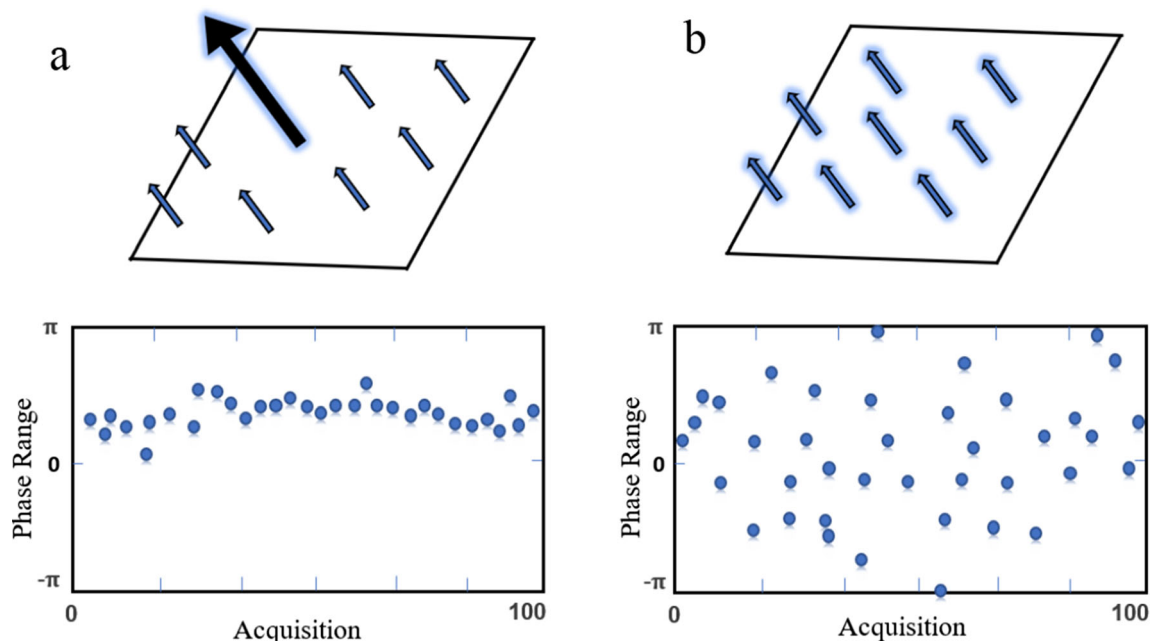


Figure 1. Phase simulations for (a) a persistent scatterer pixel and (b) a distributed scatterer pixel. The diagrams above depict the scatterers contributing to the phase of one pixel in an image, and the collocated box plots show simulations of the phase for several consecutive acquisitions. In DS, the minor scatterers are shifting randomly between each iteration. In PS, the prominent scatterer is many times brighter than the smaller juxtaposed scatterers (modified from Hooper *et al.* 2012).

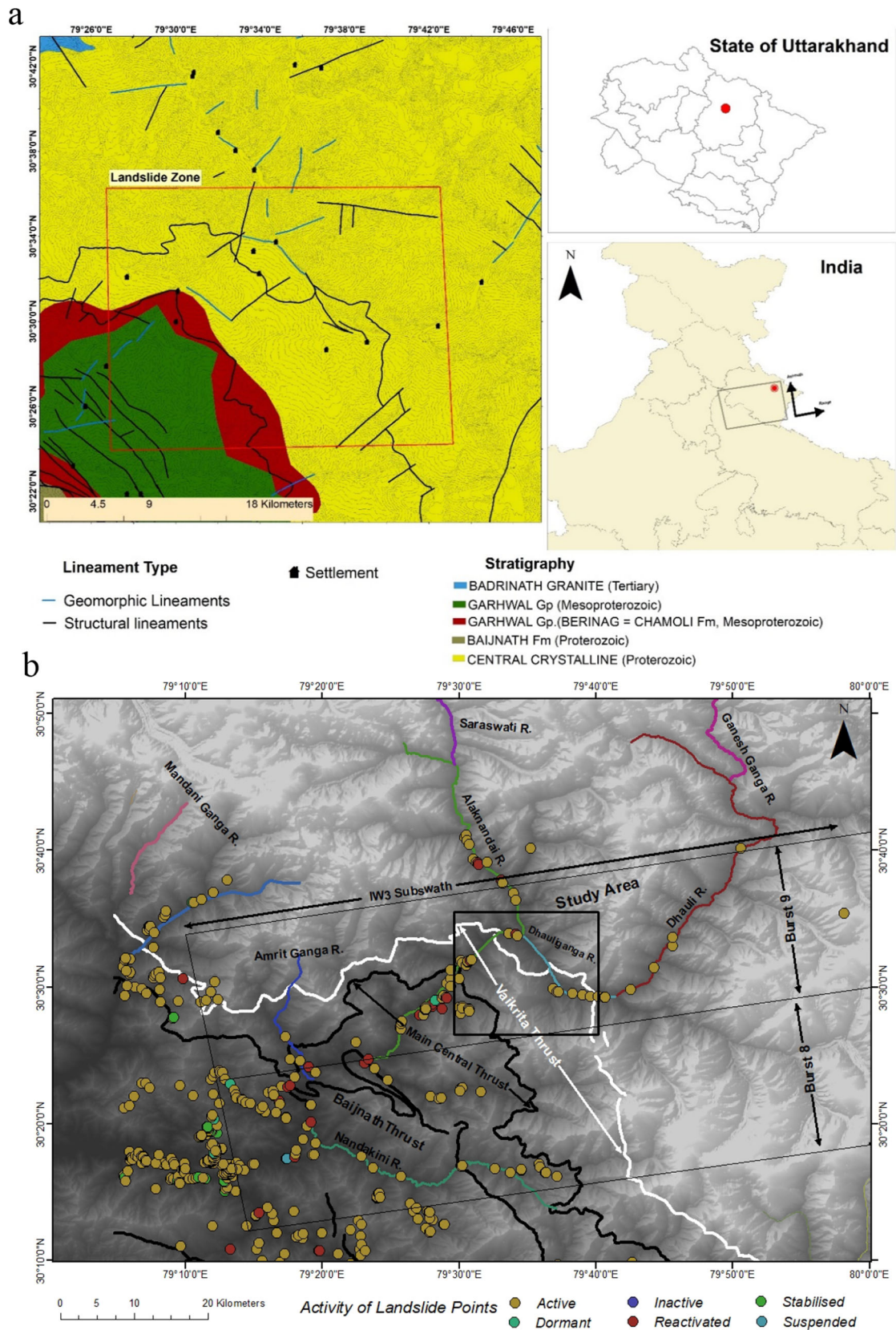


Figure 2. Geological overview of the study region. (a) Lithology of the study area, including lineaments. The inset shows the relative location of the ROI. (b) SAR burst coverage, local fluvial systems and major tectonic discontinuities. Landslide inventory map depicts the failure systematics near the recent landslide (acquired from <https://bhukosh.gsi.gov.in/>). Background is an InSAR-generated DEM.

residents with 4,500 buildings spread over 2.5 km² (BBC News 24 Jan 2023). The town lies on national highway NH58, connecting Delhi with the famous Badrinath temple, Hemkund Sahib and Mana Pass in Uttarakhand near the Indo-Tibet border. Geomorphologically, the town is located on a highly dissected hill with a structural origin. Multiple thrust systems predominate the region, especially Main Central Thrust (figure 2b), locally known as Vaikrita Thrust (Ray *et al.* 2007). The area lies in the proximal vicinity of Main Central Thrust, which has emplaced central crystalline rocks of Proterozoic age onto low-grade sedimentary successions of the Lesser Himalaya (Ray *et al.* 2007). Due to intense compressive stresses, the rocks from the underlying Joshimath Formation are thrust over the rocks of Munsiri Formation along the Vaikrita Thrust (Bisht and Rautela 2010). Munsiri Group is a highly imbricated low to medium-grade Paleoproterozoic rocks dominated by mylonitic augen gneiss (Jain *et al.* 2014). Joshimath Formation consists of medium to high-grade rocks such as gneisses, amphibolites and garniferous mica schists (figure 2a and b) with a profusion of quartz veins that manifest several tectonic features like largest ductile shear zones, progressive northward increase in metamorphic grade as well as palimpsest refolded structures (Srivastava and Tripathy 2005; Jain *et al.* 2014).

3. Materials and methods

In this study, Sentinel-1 datasets were utilized for interferogram formation and subsequent time-series extraction of PS and DS points (figure 3). Data was acquired from Alaska Satellite Facility (ASF Home Page | ASF (alaska.edu)) in the Interferometric-Wide swath mode, which is a Level-1 SLC (Single Look Complex) product through Terrain Observation by Progressive Scans (TOPS) with a 5.6-cm wavelength radar signal. Eighty-seven ascending scenes with a temporal baseline of 1056 days ranging from 21 Feb 2020 to 12 Jan 2023 were acquired from track 129 and frame 92. The temporal sampling rate in ascending data is 12 days. A master was selected for pre-processing of the data based on the approximate barycentre of spatial and temporal baseline with respect to the slave data stack to ensure maximum coherence using the latest version of ESA's (European Space Agency) SNAP (Sentinel Applications Platform). 21 July 2021 was chosen as an optimum master with bursts 7–9 (figure 4a). For PS

pre-processing, interferogram and amplitude images were generated from IW3 sub-swath in a batch mode employing GPT (Graph Processing Tool) that covered the AOI by using snap2stamps (Foumelis *et al.* 2018). Topographic phase subtraction was done with Shuttle Radar Topography Mission (SRTM) 1-arcsecond digital elevation model. Selection and extraction of PS points were done using StaMPS – Stanford Method for Persistent Scatterer (Hooper 2006; Hooper and Zebker 2007).

StaMPS (Hooper *et al.* 2004) allows the identification of PS by their characteristics of both amplitude and phase information. They usually correspond to artificial reflectors, man-made structures like buildings, pavements, or bare rock surfaces. As this approach uses a single-master strategy, it therefore produces $N - 1$ interferograms from N SLCs. For DS pre-processing, the data was run through complicated algorithms to generate PSAR and PSDS files using TomoSAR Package (Ho Tong Minh *et al.* 2020). In the DS approach ($N - 1 < M < N(N - 1)/2$), no. of M (multi-master interferograms) with short spatial-temporal baselines are generated (figure 4b and c) (Berardino *et al.* 2002). To increase the SNR (signal-to-noise ratio) of a DS, SHPs (statistically homogenous pixels) are chosen from the resolution cell by applying two-sample Kolmogorov–Smirnov test (with a certain level of significance). Pixels having a similar cumulative probability distribution (CDF) harmoniously form a brotherhood, resulting in a family of SHPs. DS candidates are selected after crossing a certain threshold number of such SHPs (Ho Tong Minh *et al.* 2020). Following the generation of DS and the phase noise estimation for the selection of PS from interferograms, both PS and DS are jointly processed inside Matlab. The joint PSDS are unwrapped, thereby yielding mean LOS deformation along with time-series displacement rate for ROI.

Toolbox for Reducing Atmospheric InSAR Noise (TRAIN) (Bekaert *et al.* 2015) was used to remove atmospheric phase screen. TRAIN uses various numerical weather model datasets to extract atmospheric signal contribution from the interferometry.

4. Results and analysis

The application of InSAR technique revealed a substantial amount of information pertaining to the landslide-induced subsidence in the Auli and

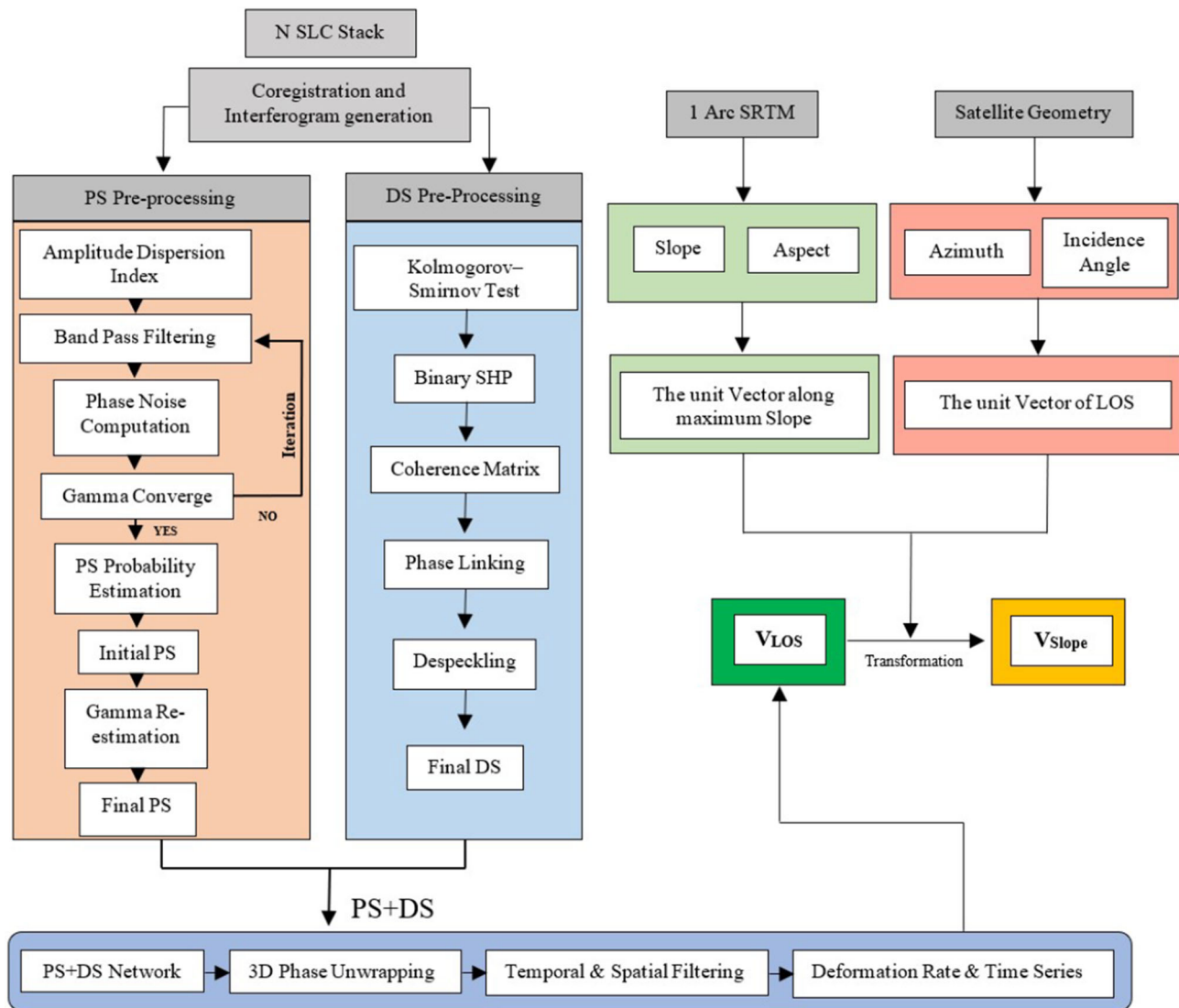


Figure 3. Flowchart showing methodology used to apply TomoSAR PSDS in this study to derive slope and LOS deformation. Both PS and DS are generated separately but processed together during filtering, unwrapping and SCLA error calculation.

Joshimath settlements. The time-series behaviour from 21 Feb 2020 to 12 January 2023, ranging from 1056 days of different radar scatterers in the 75 km² area of interest, depicts slow-moving landslides permeating the whole region. LOS velocity fields were generated by applying both Persistent Scatterer technique and PSDS technique to the affected zone. The obtained number of permanent scatterers using PSI was extremely low (6127 coherent targets) due to the dearth of civil infrastructure outside of the Joshimath along with the lack of consistent phase signal from the slow-moving slide beneath the townships of Joshimath and Auli. Profuse vegetation and incoherent soil surface led to the decreased density of 82 PS per km². The lack of sufficient PS density for phase unwrapping, deformational and vector analysis was

ameliorated by using the PSDS technique whereby coherent signals generated from distributed scatterers, especially forests, agricultural fields, bare soil and rock surfaces as well as slowly moving civil infrastructure furnishing 55,800 PSDS points with a sufficient density of 744 scatterers per km². PSDS, as compared to PSI, enhanced the point density by a factor of 10 in the observation zone. Figure 5(a and b) illustrates the mean line of sight maps whereby warm colours (orange and red) represent the movement of ground away from the satellite while green colours represent movement towards the satellite. Using the PSI approach, we identified the middle of Joshimath town as the zone with maximum displacement of −26 mm per annum and a concentric outer area of medium deformation of −15 mm per annum.

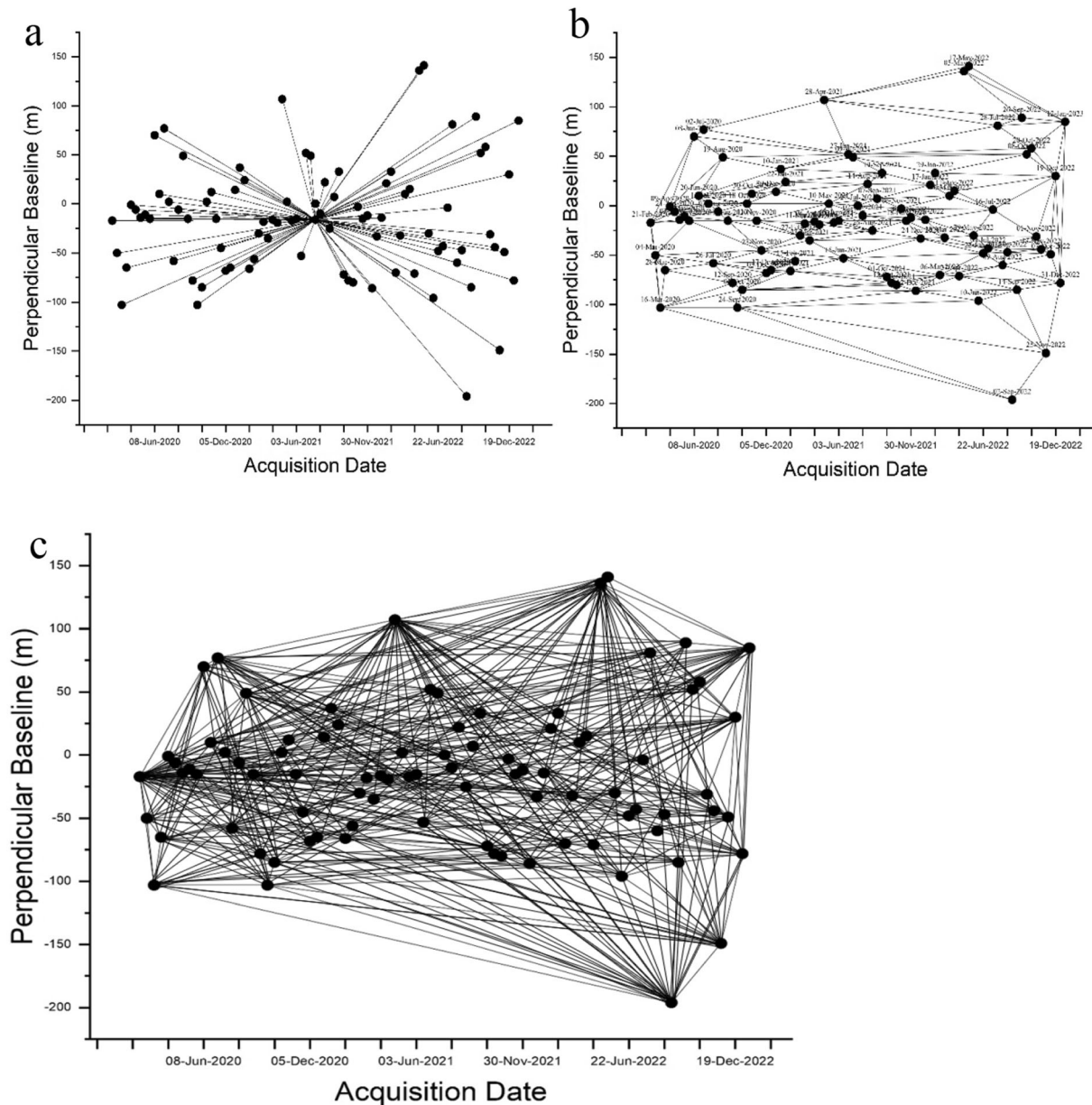


Figure 4. Temporal and spatial baseline of different InSAR approaches. Each black dot represents an SLC, while the black lines connecting these dots represent an interferogram. **(a)** Selection of appropriate master for PSI approach. **(b)** DS subset network graph. **(c)** PSDS approach with full connection network.

PSDS approach yielded a more empirical deformational field with the area of maximum deformation (-26 mm/annum) lying to the eastern side of Joshimath and the area of medium deformation (-18 mm/annum) falling within the Auli as well as middle-eastern part of the Joshimath town as shown in figure 5(a and b). Negative numbers indicate movement away from the satellite, while positive numbers indicate movement of the scatterer towards the satellite. Only a component of the ground deformation on a slope is measured by a

SAR sensor in the form of a line of sight. Therefore, we derive actual velocity along the slope (V_{slope}). Projection of LOS velocity fields into slope velocity maps is done by using the parameters of slope and satellite geometry, especially slope aspect (figure 6a), slope inclination (figure 6b), as well as the satellite heading angle (azimuth) to mark the movement of a scatterer along maximum slope plane (figure 6c).

In most translational types of displacements, the movement of the ground slide occurs along the

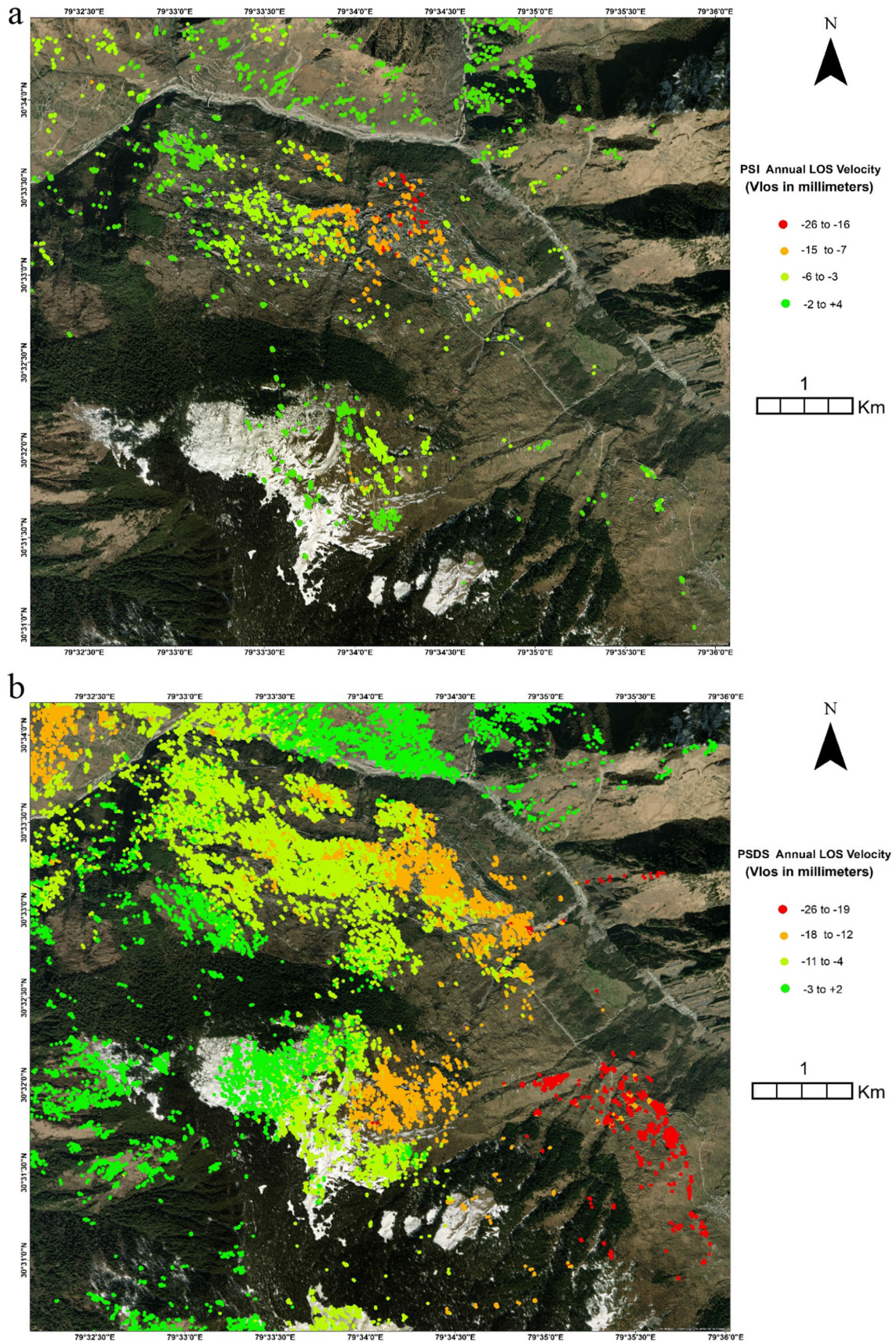


Figure 5. Mean line-of-sight (LOS) velocity for the period spanning 2020 to 2023 over Joshimath. Deformational fields as derived from (a) PS analysis and (b) PSDS analysis. Note the scarcity of points from the PS approach in rural/semi-rural setting.

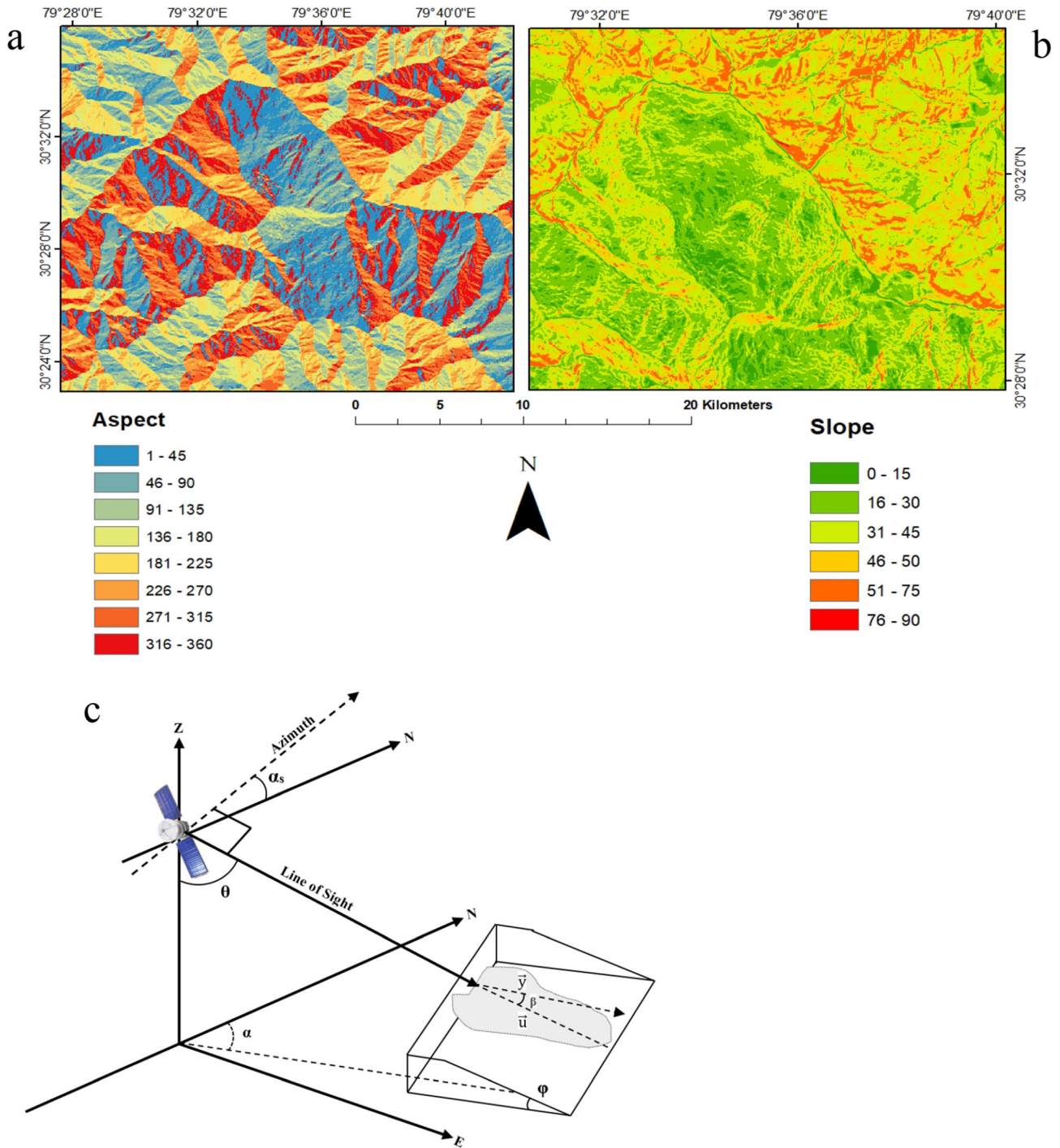


Figure 6. (a) Aspect of the Joshimath region. (b) Slope gradient and distribution in the Joshimath region. (c) Geometric relationship between the slope aspect as well as the line-of-sight direction of satellite in ascending mode. Whereas α_s is the heading angle, θ is the incidence angle, ϕ is the slope gradient, α is the slope aspect angle, $\vec{u}$ is the LOS displacement unit vector, $\vec{y}$ is the unit vector of actual slope displacement and β is the angle between these two unit vectors (modified from Dai *et al.* 2022).

direction of the steepest slope, and as such, V_{slope} is considered the perfect representative of the actual motion of the deformation. The downslope projection is generated from LOS by using following equations from Notti *et al.* (2014); Aslan *et al.* (2020); and Rehman *et al.* (2020).

$$V_{\text{slope}} = \frac{V_{\text{Los}}}{C} \quad (1)$$

The coefficient C helps in the derivation of the sensitivity of the LOS projection into the downslope direction as well as the actual portion of

the displacement as measured by the SAR sensor. This coefficient is a function of common morphological indices of slopy terrain and can be calculated by:

$$\begin{aligned} C = & N \cdot (\cos(S) \cdot \sin(A - 90^\circ)) \\ & + E \cdot (-1 \cdot (\cos(S) \cdot \cos(A - 90^\circ)) \\ & + H \cdot (\sin(S)) \end{aligned} \quad (2)$$

where S is the local slope angle, A is the aspect of study terrain with respect to the north; N , H and E are the directional cosines of the LOS vector and are calculated through the below-given equations:

$$N = \cos(90^\circ - \alpha) \cdot \cos(180^\circ - \gamma) \quad (3)$$

$$E = \cos(90^\circ - \alpha) \cdot \cos(270^\circ - \gamma) \quad (4)$$

$$H = \cos(\alpha) \quad (5)$$

whereas γ is the incidence angle of the radar waves and α is the line of sight azimuth in degrees.

Applying the above equations to the LOS velocity fields, the subsequent generation of V_{slope} maps shows a considerable amount of deformation in the Joshimath and Auli. Projection of PSI into V_{slope} (figure 7a) depicts the centre of Joshimath town with a radius of 600 m and maximum slope velocity of 84 mm/annum or 3.31 inch/year. A concentric area of radius 1500 m has subsided at the rate of 50 mm (1.97 inch) per annum. A third deformational sector with a mean slope velocity of -16 mm/year lies around the periphery of the medium deformational zone and is represented by yellow-coloured points. The green coloured dots show stable areas. PSDS approach detected a larger tract of Joshimath undergoing maximum deformation (V_{slope}) of 107 mm (4.21 inches) per annum (figure 7b) covering an area of about 5.5 km² and the area with V_{slope} of 74 mm (2.91 inches) per year in Joshimath and Auli covering an area of 3.51 and 1.34 km², respectively. The third creep zone of slow-moving landslide (yellow points) can be seen on the middle to western side of the town. A stable area represented by green dots is prevalent on the western flank of the hilly spur on which Joshimath is located.

5. Discussions

5.1 Time-series analysis

PSDS and PS interferometry generate time-series positional change of scatterers. By applying both of these techniques, four distinct deformational zones

were recognized in the Joshimath slopes represented by red, orange, yellow and green. Warm-coloured points (red, orange and yellow) show movement of scatterer patches away from the satellite, while green points show a stable area. But PSI and PSDS differ in the cognition of landslide-induced subsidence as PS interferometry cannot detect landslide to the eastern area of Joshimath where the dearth of built-up as well as profuse vegetation (figure 8) disrupts the coherence of a supposed permanent scatterer. In contrast, PSDS method is able to detect the area of maximum deformation in the far eastern sector, where the viability of PSDS method makes the detection of distributed scatterers attainable. To assess the temporal range change of different scatterers, seven 2D cross-section lines (numbering from CS1 to CS7) were drawn on the slopes of Joshimath and Auli (figures 8 and 9) in an east-to-west trend to represent the time-series change in both V_{LOS} as well as the V_{slope} of the PS/DS locality. Two-to-four points each for a CS profile were selected to analyze the evolution of such point scatterers through time. CS1, CS2 and CS4 show maximum deformational regime (figures 10, 11). Four distinct scatterer points were chosen on the CS4 to render the time-series behaviour of built-up zones in both Auli and Joshimath. The length of CS1 and CS2 is 2000 m, while the length of CS3–CS7 is 4000 m.

5.2 Groundwater level fluctuations/spring level data

The hydrogeological study carried out by the Central Ground Water Board (CGWB) in the town of Joshimath identified the presence of a discontinuous water table within the porous and permeable morainic deposits embedded with large boulders of Quarternary age (Prashant *et al.* 2023). The geological formations are moderately to steeply dipping, and the topography is frequently undulating. The groundwater is mainly drawn through handpumps. A total of four handpumps and seven monitorable springs are present in the study area (figure 12). The handpumps are drilled up to the depth of about 80 m and the zone tapped ranges from 65 to 80 m bgl, while the depth of water level in these handpumps ranges between 18 and 48 m as per the data available with Jal Sansthan, Joshimath.

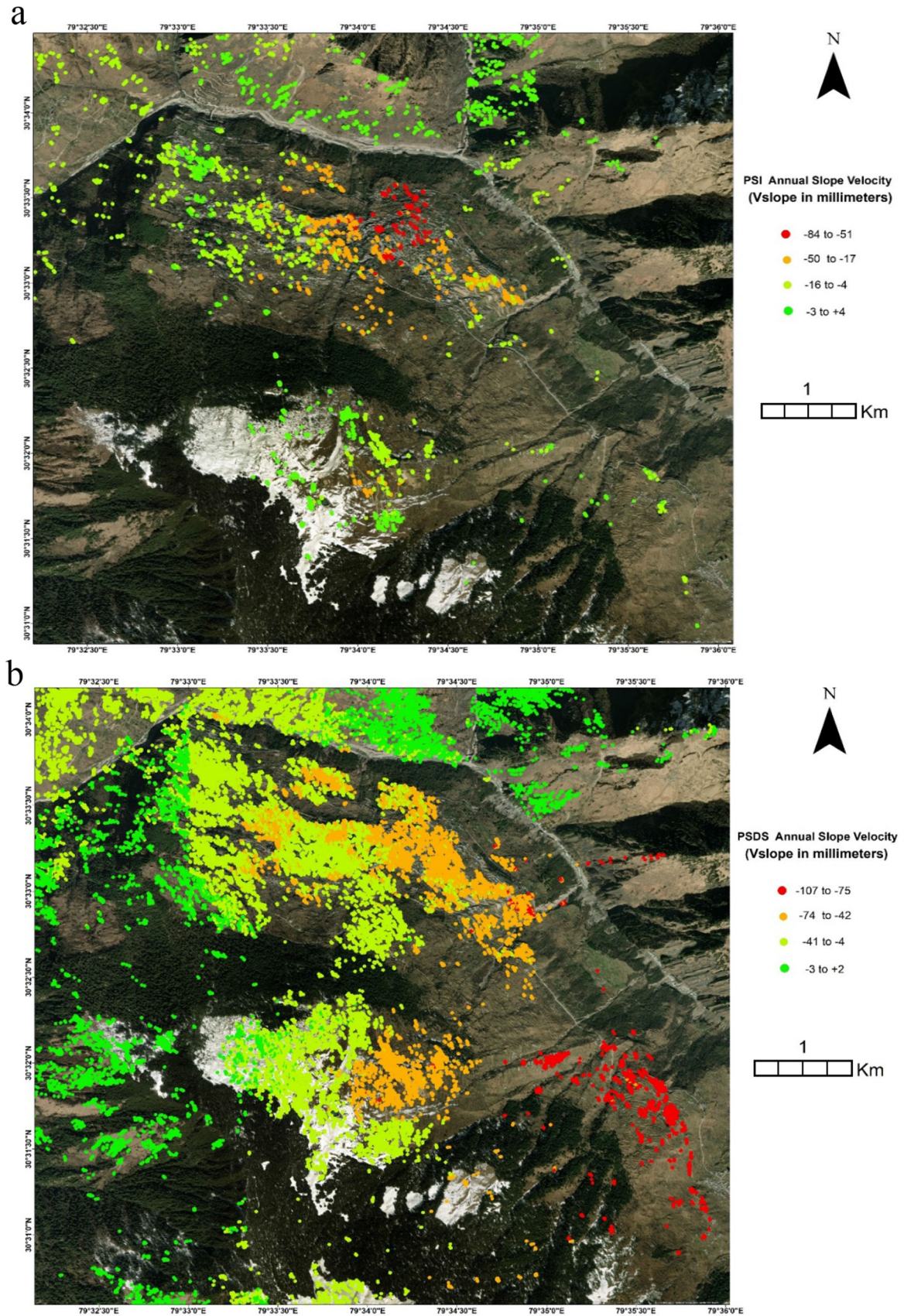


Figure 7. Mean velocity along the maximum slope direction for the period of 2020–2023 over Joshimath. Shown deformational fields are derived from the following two approaches: (a) PS analysis and (b) PSDS analysis. Note the scarcity of points from the PS approach in rural/semi-rural settings.

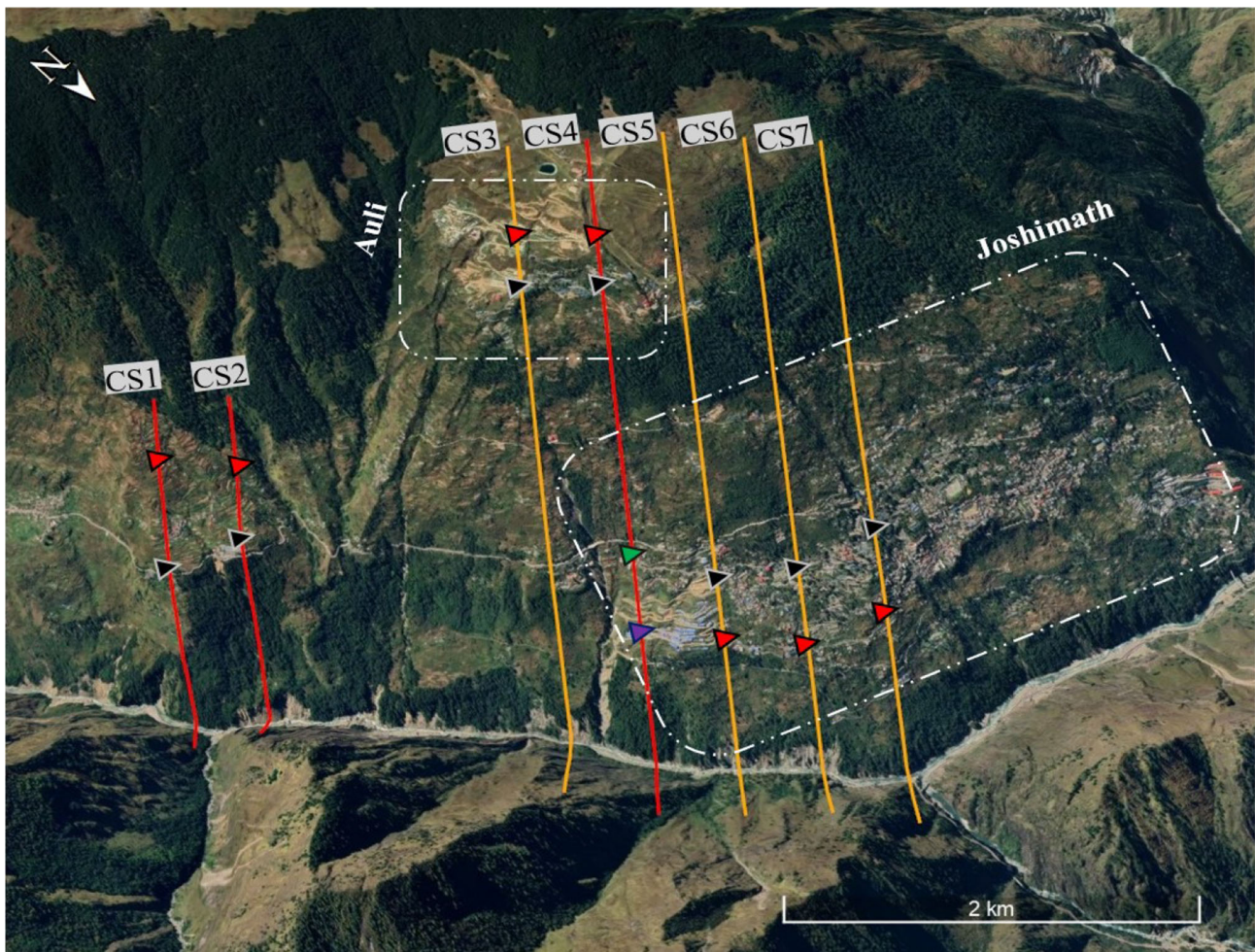


Figure 8. Satellite overlook of the Joshimath and Auli townscape with profile cross-section lines drawn over it. This study implies the deformation along CS1, CS2 and CS4 as being maximum through the observation period of 3 years.

Daily water level depth of the four handpumps has been monitored and it is observed that the water level is declining in Singhdhar area, with varying rates ranging from 20 to 60 cm per day, while in other handpumps, the water level has remained almost constant (table 1). The decline of water levels in Singhdhar handpump (figure 13) has led to the likely association of the local bearing strata with the incumbent land subsidence. The discharge of these handpumps was taken manually, ranging between 17.65 and 20.27 lpm.

Moreover, springs frequently manifest in the Joshimath region as a prevalent phenomenon. Groundwater in motion surfaces as springs at the contact of distinct lithologies through various pathways such as joints and fractures. Additionally, topographic disruptions serve as conducive sites for the formation of springs. Consequently, the majority of these springs are

depression and contact type. Discharge of these springs varies from 1 to 600 lpm, and temperature fluctuates between 12 and 13°C, as shown in table 2.

As per the CGWB report, the substantial influx of pilgrims on the Char Dham Yatra and the consistent arrival of tourists each year, numerous hotels and homestays have been built in this sensitive area to cater the accommodation requirements. Regrettably, some of these edifices have directly been constructed at the emergence points of springs or proximal to the spring zones. This has led to the disruption of spring flow, causing a cessation in the natural discharge, which in turn results in an upsurge in pore pressure, ultimately leading to groundwater bursts in the built-up of the township. In essence, the construction within spring zones may be a contributing factor to the recent occurrence of land subsidence.

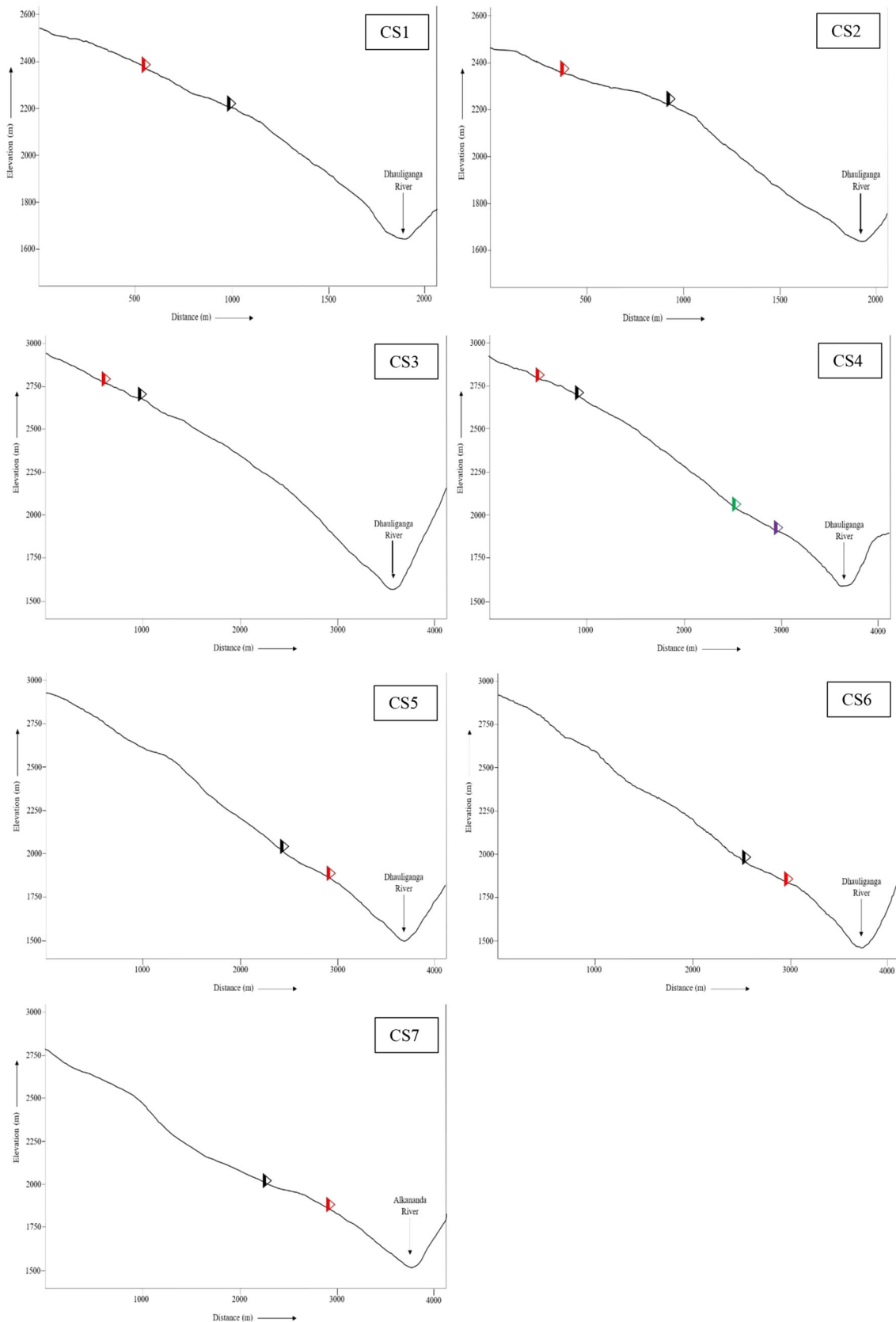


Figure 9. Cross-section profiles in east-to-west trend numbering from CS1 to CS7 were drawn illustrating downslope curves of Joshimath and Auli along with the selection of specific scatterer points to study the time-series behaviour of both LOS as well as the transformed slope velocity. 2–4 PSDS representative points were selected in the buildup zone of the slope to demonstrate their corresponding time-series evolution with respect to the satellite.

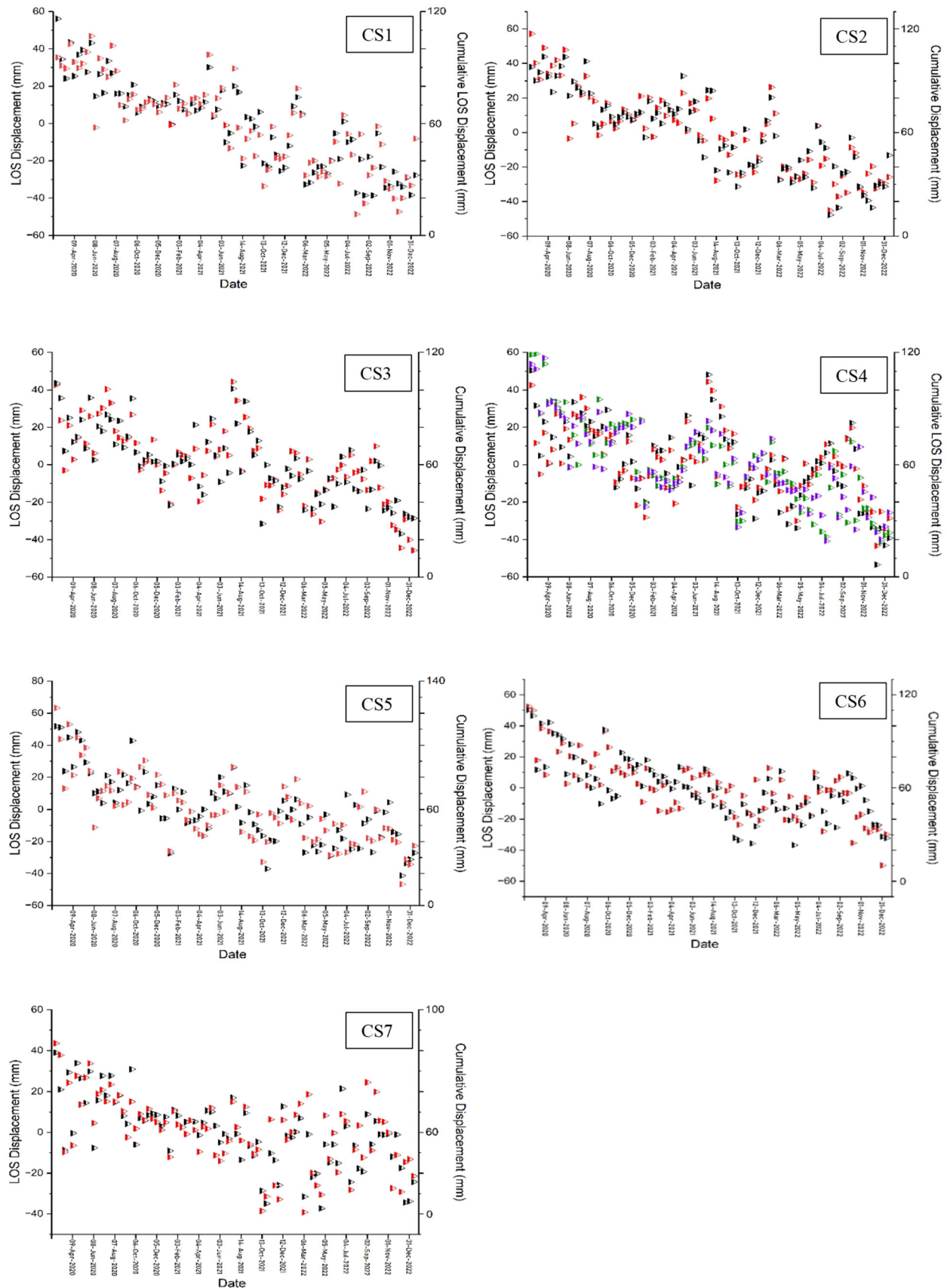


Figure 10. A view at the LOS time series of scatterers prevalent in the deformational/subsidence zone. Selected points on CS1, CS2, CS4 and CS5 show a smooth and linear trend of increased range change away from the satellite.

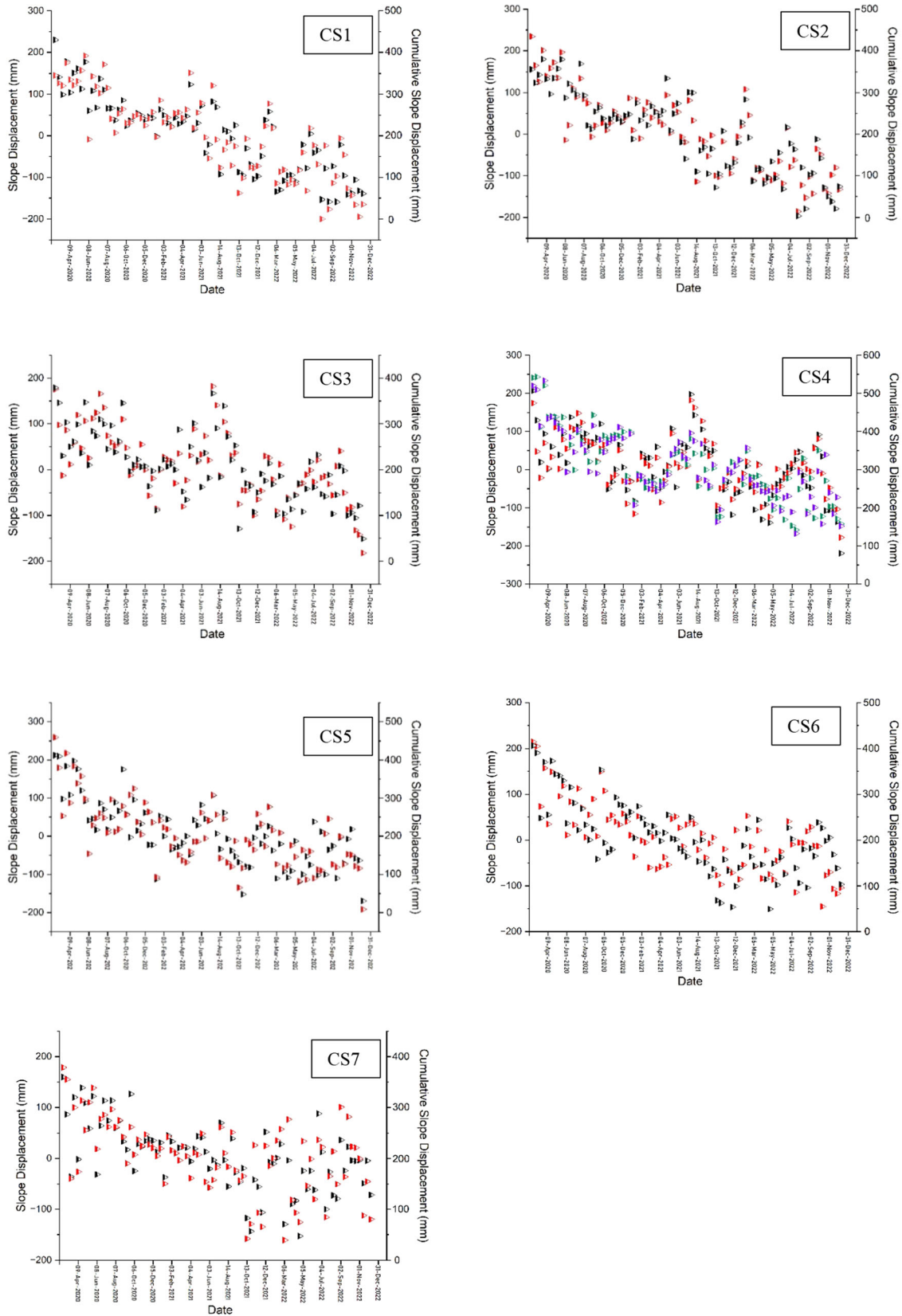


Figure 11. Time series of points on CS profiles in the direction of maximum slope. The right axis of the charts shows cumulative displacement (in mm) through temporal observation. Total cumulative displacement of CS1 and CS2 is 400 mm, CS3 is 350 mm, in CS4 (black, green and purple triangles have 400 mm for each point, red has 360 mm displacement). CS5 has 350, CS6 has 300–350 mm, and CS7 has 250–300 mm of total displacement.

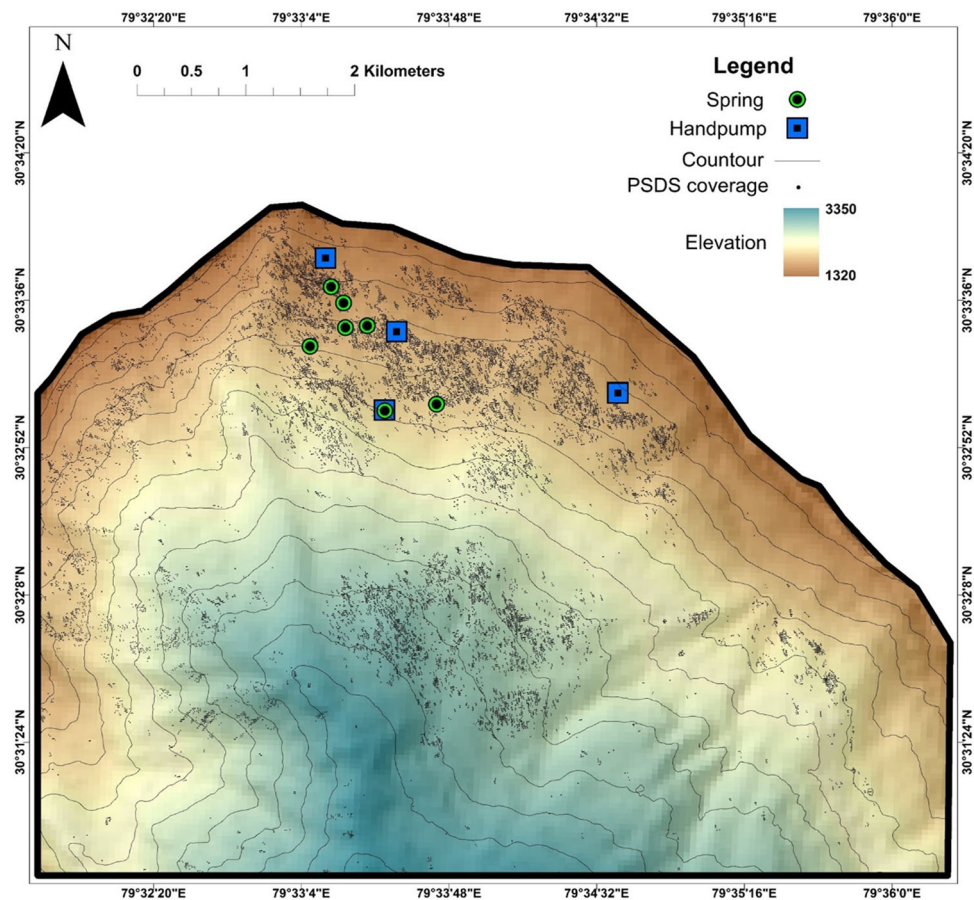


Figure 12. Location of springs, handpumps monitored by CGWB along with the coverage of our PSDS study in the Joshimath town, Chamoli district, Uttarakhand.

Table 1. Hydrogeological details of four handpumps in the study area, lpm is litres per minute.

Sl. no.	Location	Long.	Lat.	11 Jan	13 Jan	14 Jan	18 Jan	Temperature (°C)	Discharge (lpm)
1	Singhdhar	79.5590	30.5574	47.03	47.92	48.4	51.2	–	–
2	Ravigram	79.5773	30.5523	20.52	20.52	–	–	15	17.65
3	Marwari-GREFP	79.5531	30.5635	22.8	22.73	–	–	15	–
4	Sunil	79.5580	30.5509	18.37	18.22	18.15	–	13	20.27

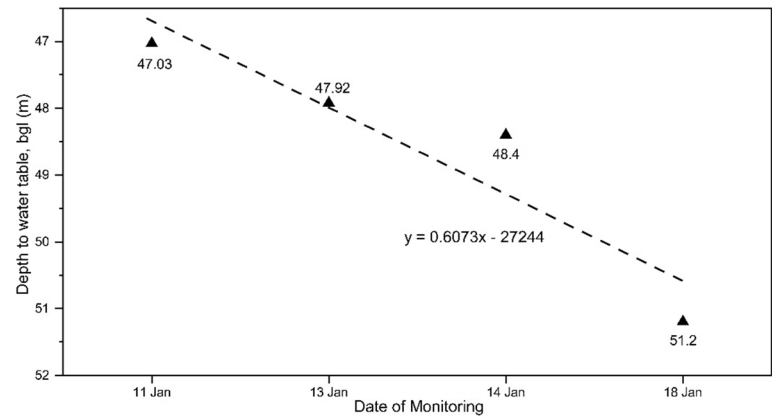


Figure 13. Hydrograph of Singhdhar handpump (figure 12 and table 1) showing a steady decrease of the water table with respect to the ground level.

Table 2. Details of discharge and temperature of monitored springs in and around Joshimath Town.

Sl. no.	Location	Long.	Lat.	Discharge (lpm)	Temp. (°C)	Remarks
1	JPVL-Marvadi	79.55356	30.56113	250-60	13	Badminton Hall-Inside Campus
2	Nauganga	79.56227	30.55141	600	12	Sunil-Jal Sansthan Water supply tank
3	Sunil	79.55802	30.55085	1	13	Near ITBP Campus
4	Dronagiri	79.55183	30.55618	35.8	13	–
5	Singhdhar	79.55475	30.55773	28.91	13	Near Bhatt House
6	Lower-Singhdhar	79.5546	30.55978	–	12.5	Just above JPVL
7	Singhdhar–Gopaldhar	79.55656	30.55789	35.29	13	Near Panchvati Hotel

6. Conclusion

In this study, we used radar imagery to perform both PS and PSDS interferometry in view of recent landslide-induced subsidence in Joshimath in the State of Uttarakhand, India, to measure its extent as well as gauge the amount of deformation. Interferometric stack was acquired from ascending orbit of Sentinel-1 from 21 Feb 2020 up to 12 January 2023 with a temporal range of 1056 days covering an area of 75 km². Both PS and PSDS approaches yielded the LOS velocity fields on the slopes of Joshimath and Auli, which were subsequently projected onto the velocity along the maximum slope (V_{slope}) using appropriate methods of vector transformation as shown in equations (1 and 2). Using PSI technique (figure 7a), we could discern different deformational zones with insufficient point density (82 points per km²) owing to the semi-rural setup of the township. PSDS approach led to the cognition of four distinct deformational regimes represented by different colour schemes (figure 7b) from phase unwrapping of high-coherent and high-density target points (744 scatterers per km²). To the eastern and middle-eastern sector, the zone of maximum deformation (red dots) covering an area of about 5.5 km², undergoing 107 mm (4.21 inches) per annum was detected. Zone of medium deformation with a displacement of 74 mm (2.91 inches) per year was revealed, covering an area of 3.51 km² in Joshimath and 1.34 km² in Auli. A latent zone of slow-moving landslide (yellow points) is evident from the signal analysis, which can be interpreted as a soil creep. Seven cross-sectional lines illustrating 2D profiles incorporating the nearby valley floor were drawn on the subsidence-affected slope. Two to four scatterer targets from the built-up zone were

selected on these CS lines to monitor and analyze their temporal evolution with respect to their location on the CS profile. Target systematics in time-series plots show scatterers on CS1, CS2 and CS4 have a cumulative displacement of 400 mm or 15.75 inches in the last 3 years. CS3 and CS5 each have a cumulative displacement of about 350 mm or 13.78 inches. Points on CS6 have a cumulative displacement of 300 mm (11.82 inches) to 350 mm (13.78 inches), and scatterers on CS7 have a net displacement of 250 mm (9.85 inches) to 300 mm (11.82 inches). As such, our study in the region has highlighted the prospects of applying novel PSDS time-series InSAR to geodetic monitoring and the potential of progressive failure in such landslide events. Proper forecasting of meta-stable slopes and a thorough investigation of deformational extent play an important role in the prevention of catastrophic failures corresponding to the effect of slope cutting, improper drainage, tunnelling and civil infrastructure built-up in fragile ecosystems like that of Joshimath.

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Author statement

Asrar Ahmad Rather: Original manuscript drafting, conceptualization, methodology, analysis and interpretation. Syed Kaiser Bukhari: Formal visualization, supervision, review and editing.

References

- 900 members of 29 families displaced in Joshimath land subsidence, says State Disaster Management Secretary – The Hindu; <https://www.thehindu.com/news/national/other-states/900-members-of-29-families-displaced-in-joshimath-land-subsidence-says-state-disaster-management-secretary/article66413601.ece>.
- Agarwal S, Kumar V, Kumar S, Sundriyal Y, Bagri D S, Chauhan N, Kaushik S, Khan F and Rana N 2023 Identifying potential hotspots of land use/land cover change in the last 3 decades, Uttarakhand, NW Himalaya.
- Aslan G, Fomelis M, Raucoules D, De Michele M, Bernardie S and Cakir Z 2020 Landslide mapping and monitoring using persistent scatterer interferometry (PSI) technique in the French Alps; *Rem. Sens.* **12**(8) 1305.
- Awasthi A K, Tabatabaei S H, Singh B and Mehrotra G S 1995 Terrain attributes and drainage texture as indicators of landslide occurrence in a part of Garhwal Himalaya, India; *J. Nepal Geol. Soc.* **11** 289–298.
- Barnard P L, Owen L A, Sharma M C and Finkel R C 2001 Natural and human-induced landsliding in the Garhwal Himalaya of northern India; *Geomorphology* **40**(1–2) 21–35.
- Bekaert D P, Walters R J, Wright T J, Hooper A J and Parker D J 2015 Statistical comparison of InSAR tropospheric correction techniques; *Rem. Sen. Environ.* **170** 40–47, <https://doi.org/10.1016/J.RSE.2015.08.035>.
- Bekaert D P, Handwerger A L, Agram P and Kirschbaum D B 2020 InSAR-based detection method for mapping and monitoring slow-moving landslides in remote regions with steep and mountainous terrain: An application to Nepal; *Rem. Sen. Environ.* **249** 111983.
- Berardino P, Fornaro G, Lanari R and Sansosti E 2002 A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms; *IEEE Trans. Geosci. Rem. Sens.* **40**(11) 2375–2383, <https://doi.org/10.1109/TGRS.2002.803792>.
- Bisht M P and Rautela P 2010 Disaster looms large over Joshimath; *Curr. Sci.* **98**(10) 1271.
- Bürgmann R, Rosen P A and Fielding E J 2000 Synthetic aperture radar interferometry to measure Earth's surface topography and its deformation; *Ann. Rev. Earth Planet. Sci.* **28**(1) 169–209, <https://doi.org/10.1146/annurev.earth.28.1.169>.
- Chadha R K 2023 2022 Joshimath subsidence; *J. Geol. Soc. India* **99**(2).
- Chauhan S, Sharma M and Arora M K 2010 Landslide susceptibility zonation of the Chamoli region, Garhwal Himalayas, using logistic regression model; *Landslides* **7** 411–423, <https://doi.org/10.1007/s10346-010-0202-3>.
- Chen L, Liu P, Liao W, Zhang K, He F and Zeng L 2020 Regional-scale detection of unrecorded landslides in mountainous terrains by using interferometric stacking technique; *IOP Con. Series: Earth Environ. Sci.* **570**(4), <https://doi.org/10.1088/1755-1315/570/4/042052>.
- Dai Keren, Deng Jin, Xu Qiang, Li Zhenhong Li, Shi Xianlin Shi, Hancock Craig, Wen Ningling, Zhang Lele and Zhou Guanchen 2022 Interpretation and sensitivity analysis of the InSAR line of sight displacements in landslide measurement; *GISci. Rem. Sens.* **59**(1) 1226–1242.
- Devrani R and Singh V 2014 Evolution of valley-fill terraces in the Alaknanda Valley, NW Himalaya: Its implication on river response studies; *Geomorphology* **227** 112–122, <https://doi.org/10.1016/j.geomorph.2014.04.030>.
- Dong J, Zhang L, Tang M, Liao M, Xu Q, Gong J and Ao M 2018 Mapping landslide surface displacements with time series SAR interferometry by combining persistent and distributed scatterers: A case study of Jiaju landslide in Danba, China; *Rem. Sens. Environ.* **205** 180–198, <https://doi.org/10.1016/j.rse.2017.11.022>.
- Even M and Schulz K 2018 InSAR deformation analysis with distributed scatterers: A review complemented by new advances; *Rem. Sens.* **10**(5) 744, <https://doi.org/10.3390/rs10050744>.
- Ferretti A, Prati C and Rocca F 2001 Permanent scatterers in SAR interferometry; *IEEE Trans. Geosci. Rem. Sens.* **39**(1) 8–20.
- Fiorucci F, Cardinali M, Carlà R, Rossi M, Mondini A C, Santurri L, Ardizzone F and Guzzetti F 2011 Seasonal landslide mapping and estimation of landslide mobilization rates using aerial and satellite images; *Geomorphology* **129**(1–2) 59–70, <https://doi.org/10.1016/j.geomorph.2011.01.013>.
- Fomelis M, Blasco J M, Desnos Y L, Engdahl M, Fernández D, Veci L, Lu J and Wong C 2018 ESA SNAP-StaMPS integrated processing for Sentinel-1 persistent scatterer interferometry; *IGARSS 2018-2018 IEEE Intern. Geosci. and Rem. Sens. Symp.*, pp. 1364–1367.
- Fresh cracks appear on road between Joshimath and Badrinath ahead of Char Dham Yatra – The Hindu; <https://www.thehindu.com/news/national/other-states/fresh-cracks-appear-on-road-between-joshimath-and-badrinath-ahead-of-char-dham-yatra/article66532181.ece>.
- Gerrard J 1994 The landslide hazard in the Himalayas: Geological control and human action; *Geomorphology* **10** 221–230.
- Hanssen R 2001 *Radar interferometry: Data interpretation and error analysis*; Vol. 2, Springer Science and Business Media.
- Ho Tong Minh D and Ngo Y N 2022 Compressed SAR interferometry in the big data era; *Rem. Sens.* **14**(2) 390, <https://doi.org/10.3390/rs14020390>.
- Ho Tong Minh D, NGO Y, Lê T T, Le T C, Bui H S, Vuong Q V and Le Toan T 2020 Quantifying horizontal and vertical movements in Ho Chi Minh City by Sentinel-1 Radar Interferometry; Preprints 2020, 2020120382.
- Hooper A 2006 Persistent scatterer radar interferometry for crustal deformation studies and modeling of volcanic deformation; *J. Geol. Res.* **112** B07407.
- Hooper A and Zebker H 2007 Phase unwrapping in three dimensions with application to InSAR time series; *J. Opt. Soc. Am.* **24**(9) 2737–2747.

- Hooper A, Zebker H, Segall P and Kampes B 2004 A new method for measuring deformation on volcanoes and other natural terrains using InSAR persistent scatterers; *Geol. Res. Lett.* **31**(23), <https://doi.org/10.1029/2004GL021737>.
- Hooper A, Bekaert D, Spaans K and Arian M 2012 Recent advances in SAR interferometry time series analysis for measuring crustal deformation; *Tectonophys.* **514** 1–3, <https://doi.org/10.1016/j.tecto.2011.10.013>.
- Hussain M A, Chen Z, Wang R and Shoaib M 2021 PS-InSAR-based validated landslide susceptibility mapping along Karakorum highway, Pakistan; *Rem. Sen.* **13**(20), <https://doi.org/10.3390/rs13204129>.
- Ives Jack D and Bruno M 2001 *The Himalayan dilemma: Reconciling development and conservation*; Psychol. Press, Routledge, 329p.
- Jain A K, Shreshtha M, Seth P, Kanyal L, Carosi R, Montomoli C, Iaccarino S, Mukherjee P K, Law R, Singh S and Rai S M 2014 The Higher Himalayan Crystallines, Alaknanda–Dhaulī Ganga Valleys, Garhwal Himalaya, India; *J. Virtual Expl.* **47** 8.
- Jennifer J J, Saravanan S and Pradhan B 2022 Persistent scatterer interferometry in the post-event monitoring of the Idukki landslides; *Geocarto Int.* **37**(5) 1514–1528, <https://doi.org/10.1080/10106049.2020.1778101>.
- Joshimath sinking? Panic grips Uttarakhand town as houses develop deep cracks | The Indian Express; <https://indianexpress.com/photos/india-news/uttarakhand-town-joshimath-houses-develop-deep-cracks-8366102/>.
- Joshimath: What's the future of India's sinking Himalayan town? – BBC News; <https://www.bbc.com/news/world-asia-india-64369752>.
- Kanungo D P and Sharma S 2014 Rainfall thresholds for prediction of shallow landslides around Chamoli–Joshimath region, Garhwal Himalayas, India; *Landslides* **11**(4) 629–638, <https://doi.org/10.1007/s10346-013-0438-9>.
- Lacroix P, Gavillon T, Bouchant C, Lavé J, Mugnier J L, Dhungel S and Vernier F 2022 SAR and optical images correlation illuminates post-seismic landslide motion after the Mw 7.8 Gorkha earthquake (Nepal); *Sci. Rep.* **12**(1) 6266, <https://doi.org/10.1038/s41598-022-10016-2>.
- Lundgren P, Usai S, Sansosti E, Lanari R, Tesauero M, Fornaro G and Berardino P 2001 Modeling surface deformation observed with synthetic aperture radar interferometry at Campi Flegrei caldera; *J. Geol. Res. Solid Earth* **106**(B9) 19355–19366, <https://doi.org/10.1029/2001jb000194>.
- Mey J, Guntu R K, Plakias A, Silva de Almeida I and Schwanghart W 2023 More than one landslide per road kilometre – surveying and modelling mass movements along the Rishikesh–Joshimath (NH-7) highway, Uttarakhand, India; *Nat. Haz. Earth Syst. Sci. Discuss.*, <https://doi.org/10.5194/nhess-2022-295>.
- Nawani P C 2015 Groundwater ingress in head race tunnel of Tapovan: Vishnugad hydroelectric project in Higher Himalaya, India; *Engineering Geology for Society and Territory – Volume 6, Applied Geology for Major Engineering Projects*, pp. 941–945, https://doi.org/10.1007/978-3-319-09060-3_171.
- Notti D, Herrera G, Bianchini S, Meisina C, García-Davalillo J C and Zucca F 2014 A methodology for improving landslide PSI data analysis; *Int. J. Rem. Sens.* **35**(6) 2186–2214, <https://doi.org/10.1080/01431161.2014.889864>.
- Panikkar S V and Subramanyan V 1996 A geomorphic evaluation of the landslides around Dehradun and Mussoorie, Uttar Pradesh, India; *Geomorphology* **15**(2) 169–181.
- Prashant R, Tomar V, Singh S and Aniruddh Singh 2023 A hydrogeological investigation report on land subsidence in Joshimath Town, Chamoli District, Uttarakhand; CGWB, Dep. of Water Resour., River Dev. and Ganga Rejuv., Min. of Jal Shakti, Govt of India, Dehradun.
- Ray L, Bhattacharya A and Roy S 2007 Thermal conductivity of higher Himalayan crystallines from Garhwal Himalaya, India; *Tectonophys.* **434**(1–4) 71–79, <https://doi.org/10.1016/j.tecto.2007.02.003>.
- Rehman M U, Zhang Y, Meng X, Su X, Catani F, Rehman G, Yue D, Khalid Z, Ahmad S and Ahmad I 2020 Analysis of landslide movements using interferometric synthetic aperture radar: A case study in Hunza–Nagar Valley, Pakistan; *Rem. Sens.* **12**(12) 2054, <https://doi.org/10.3390/RS12122054>.
- Roy P, Martha T R, Khanna K, Jain N and Kumar K V 2022 Time and path prediction of landslides using InSAR and flow model; *Rem. Sens. Environ.* **271** 112899, <https://doi.org/10.1016/j.rse.2022.112899>.
- Saha A K, Gupta R P and Arora M K 2002 GIS-based landslide hazard zonation in the Bhagirathi (Ganga) valley, Himalayas; *Int. J. Rem. Sens.* **23**(2) 357–369, <https://doi.org/10.1080/01431160010014260>.
- Sati S P and Gahalaut V K 2013 The fury of the floods in the north-west Himalayan region: The Kedarnath tragedy; *Geom. Nat. Haz. Risk* **4**(3) 193–201, <https://doi.org/10.1080/19475705.2013.827135>.
- Siddique T, Haris P M and Pradhan S P 2022 Unraveling the geological and meteorological interplay during the 2021 Chamoli disaster, India; *Nat. Haz. Res.* **2**(2) 75–83, <https://doi.org/10.1016/j.nhres.2022.04.003>.
- Srivastava H B and Tripathy N R 2005 Shear zone structures from the Main Central Thrust Zone of the Joshimath area, Garhwal Himalaya; *Paleont. Soc. India, Spec. Publ.* **2** 53–64.
- Uttarakhand cabinet approves compensation policy for affected families in Joshimath | India News, The Indian Express; <https://indianexpress.com/article/india/uttarakhand-cabinet-approves-compensation-policy-for-affected-families-in-joshimath-8447482/>.
- Verma S, Sharma A, Yadava P K, Gupta P, Singh J and Payra S 2022 Rapid flash flood calamity in Chamoli, Uttarakhand region during Feb 2021: An analysis based on satellite data; *Nat. Haz.* **112**(2) 1379–1393, <https://doi.org/10.1007/s11069-022-05232-y>.
- Vöge M, Frauenfelder R, Ekseth K, Arora M K, Bhattacharya A and Basin R K 2015 The use of SAR interferometry for landslide mapping in the Indian Himalayas; *Int. Arch. Photog. Rem. Sens. Spat. Inf. Sci.* **40** 857–863, <https://doi.org/10.5194/isprsarchives-XL-7-W3-857-2015>.
- Yao J, Yao X and Liu X 2022 Landslide detection and mapping based on SBAS-InSAR and PS-InSAR: A case study in Gongjue County, Tibet, China; *Rem. Sens.* **14**(19) 4728, <https://doi.org/10.3390/rs14194728>.
- Zhang L, Dai K, Deng J, Ge D, Liang R, Li W and Xu Q 2021 Identifying potential landslides by stacking-InSAR in southwestern China and its performance comparison with SBAS-InSAR; *Rem. Sens.* **13**(18) 3662, <https://doi.org/10.3390/rs13183662>.

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